

Technical Documentation

Wind Turbine Generator Systems

3.8-130 - 50/60 Hz



Calculated Power Curve and Thrust Coefficient

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Table of Contents

1 Calculated Power Curve.....5

2 Power Curves for different Air Density and Turbulence Intensity7

3 Applicability10

4 Cut-Out and Re-Cut-In Wind Speeds.....12

5 Calculated Thrust Coefficient.....13

6 Guidelines for Pre-construction Energy Assessment.....14

1 Calculated Power Curve

Wind turbine type: 3.8-130
 Grid standard: 50 Hz and 60 Hz
 Rotor diameter: 130 m

The calculated power curve at an average air density of 1.225 kg/m³ is listed in Table 1. This power curve is consistent with the power curve definition in the IEC 61400-12-1 (2005) for hub height wind speed.

Turbulence intensity (TI) ranges are defined in section 3. The method for selecting the appropriate turbulence intensity is provided in section 6.

Wind Speed at Hub Height [m/s]	Electrical Power [kW] with			C _{p,e} Medium TI
	Medium TI Band	Low TI Band	High TI Band	
3.0	32	20	62	0.14
3.5	105	93	139	0.30
4.0	198	184	235	0.38
4.5	309	295	350	0.42
5.0	442	426	487	0.43
5.5	600	582	654	0.44
6.0	787	765	849	0.45
6.5	1007	982	1074	0.45
7.0	1261	1233	1337	0.45
7.5	1550	1522	1617	0.45
8.0	1869	1847	1924	0.45
8.5	2203	2191	2238	0.44
9.0	2534	2532	2544	0.43
9.5	2843	2848	2830	0.41
10.0	3131	3141	3081	0.39
10.5	3370	3403	3285	0.36
11.0	3558	3604	3467	0.33
11.5	3694	3738	3602	0.30
12.0	3764	3798	3685	0.27
12.5	3802	3823	3741	0.24
13.0	3821	3830	3774	0.21
13.5	3826	3830	3798	0.19
14.0	3829	3830	3811	0.17
14.5	3830	3830	3819	0.15
15.0	3830	3830	3823	0.14
15.5	3830	3830	3823	0.13
16.0	3830	3830	3825	0.12
16.5	3830	3830	3826	0.10
17.0	3830	3830	3830	0.10
17.5	3830	3830	3830	0.09
18.0	3830	3830	3830	0.08
18.5	3830	3830	3830	0.07
19.0	3830	3830	3830	0.07
19.5	3830	3830	3830	0.06

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Wind Speed at Hub Height [m/s]	Electrical Power [kW] with			C _{p,e} Medium TI
	Medium TI Band	Low TI Band	High TI Band	
20.0	3830	3830	3830	0.06
20.5	3830	3830	3830	0.05
21.0	3830	3830	3830	0.05
21.5	3830	3830	3830	0.05
22.0	3830	3830	3830	0.04
22.5	3830	3830	3830	0.04
23.0	3830	3830	3830	0.04
23.5	3830	3830	3830	0.04
24.0	3830	3830	3830	0.03
24.5	3830	3830	3830	0.03
25.0	3830	3830	3830	0.03

Table 1: Calculated power curve table as a function of hub height wind speed for standard air density

2 Power Curves for different Air Density and Turbulence Intensity

Following tables show the calculated power curves for different air density values and turbulence intensity classes. The excel file embedded with this document contains data for a larger range of air densities than in the tables below.

Wind Speed at Hub Height [m/s]	Electrical Power [kW]										
	Air density $\rho = 1.02$	Air density $\rho = 1.04$	Air density $\rho = 1.06$	Air density $\rho = 1.08$	Air density $\rho = 1.1$	Air density $\rho = 1.12$	Air density $\rho = 1.14$	Air density $\rho = 1.16$	Air density $\rho = 1.18$	Air density $\rho = 1.2$	Air density $\rho = 1.225$
	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³
3.0	18	19	20	22	23	25	26	27	29	30	32
3.5	79	82	84	87	89	92	94	97	99	102	105
4.0	156	160	164	168	172	176	180	184	188	193	198
4.5	249	255	260	266	272	278	284	290	296	302	309
5.0	359	367	375	383	392	400	408	416	424	432	442
5.5	492	502	513	523	534	545	555	566	577	587	600
6.0	646	660	674	687	701	715	728	742	756	770	787
6.5	829	846	864	881	898	916	933	950	968	985	1007
7.0	1042	1063	1085	1106	1128	1149	1171	1192	1214	1235	1261
7.5	1286	1312	1339	1365	1391	1417	1442	1468	1493	1519	1550
8.0	1561	1593	1624	1656	1687	1716	1746	1775	1805	1834	1869
8.5	1865	1901	1938	1974	2011	2042	2074	2105	2137	2168	2203
9.0	2183	2224	2265	2306	2347	2378	2409	2439	2470	2501	2534
9.5	2501	2545	2589	2633	2677	2705	2732	2760	2787	2815	2843
10.0	2804	2850	2896	2942	2988	3012	3036	3060	3084	3108	3131
10.5	3086	3130	3173	3217	3260	3279	3297	3316	3334	3353	3370
11.0	3323	3361	3398	3436	3474	3488	3502	3517	3531	3545	3558
11.5	3528	3556	3584	3612	3640	3649	3658	3668	3677	3686	3694
12.0	3660	3678	3697	3715	3734	3739	3744	3750	3755	3760	3764
12.5	3742	3753	3764	3775	3786	3789	3792	3794	3797	3800	3802
13.0	3785	3792	3798	3805	3812	3813	3815	3816	3818	3819	3821
13.5	3811	3814	3816	3819	3822	3823	3824	3824	3825	3826	3826
14.0	3821	3823	3824	3826	3827	3827	3828	3828	3829	3829	3829
14.5	3827	3827	3828	3828	3829	3829	3829	3830	3830	3830	3830
15.0	3829	3829	3830	3830	3830	3830	3830	3830	3830	3830	3830
15.5 to cutout ¹	3830	3830	3830	3830	3830	3830	3830	3830	3830	3830	3830

Table 2: Calculated power curve for different values of the air density for medium turbulence intensity

¹ For more details on the cut-out behaviour visit section 3 "Cut-Out and Re-Cut-In Wind Speeds"

Wind Speed at Hub Height [m/s]	Electrical Power [kW]										
	Air density $\rho = 1.02$	Air density $\rho = 1.04$	Air density $\rho = 1.06$	Air density $\rho = 1.08$	Air density $\rho = 1.1$	Air density $\rho = 1.12$	Air density $\rho = 1.14$	Air density $\rho = 1.16$	Air density $\rho = 1.18$	Air density $\rho = 1.2$	Air density $\rho = 1.225$
	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³
3.0	8	9	10	12	13	14	15	16	17	19	20
3.5	69	71	74	76	78	81	83	86	88	90	93
4.0	145	148	152	156	160	164	168	172	175	179	184
4.5	237	242	248	254	259	265	271	276	282	288	295
5.0	346	354	362	370	377	385	393	401	409	416	426
5.5	476	486	497	507	517	528	538	548	559	569	582
6.0	628	642	655	668	682	695	708	722	735	748	765
6.5	808	825	842	859	876	893	910	927	944	961	982
7.0	1018	1039	1060	1081	1102	1123	1144	1165	1186	1207	1233
7.5	1259	1285	1310	1336	1362	1388	1413	1439	1464	1490	1522
8.0	1535	1566	1598	1629	1660	1690	1720	1750	1780	1810	1847
8.5	1843	1880	1916	1953	1990	2023	2056	2088	2121	2154	2191
9.0	2171	2213	2254	2296	2338	2370	2402	2434	2466	2498	2532
9.5	2498	2544	2589	2635	2680	2708	2736	2763	2791	2819	2848
10.0	2808	2855	2903	2950	2998	3022	3046	3069	3093	3117	3141
10.5	3097	3144	3192	3239	3287	3306	3326	3345	3365	3384	3403
11.0	3353	3394	3436	3477	3519	3534	3548	3563	3577	3592	3604
11.5	3570	3600	3631	3661	3691	3699	3707	3716	3724	3732	3738
12.0	3708	3725	3742	3759	3776	3780	3784	3788	3792	3796	3798
12.5	3780	3789	3797	3806	3814	3815	3817	3818	3820	3821	3823
13.0	3812	3816	3819	3823	3826	3827	3828	3828	3829	3830	3830
13.5	3825	3826	3827	3828	3829	3829	3829	3830	3830	3830	3830
14.0	3828	3829	3829	3830	3830	3830	3830	3830	3830	3830	3830
14.5 to cutout ²	3830	3830	3830	3830	3830	3830	3830	3830	3830	3830	3830

Table 3: Calculated power curve for different values of the air density for low turbulence intensity

² For more details on the cut-out behaviour visit section 3 "Cut-Out and Re-Cut-In Wind Speeds"

Wind Speed at Hub Height [m/s]	Electrical Power [kW]										
	Air density $\rho = 1.02$	Air density $\rho = 1.04$	Air density $\rho = 1.06$	Air density $\rho = 1.08$	Air density $\rho = 1.1$	Air density $\rho = 1.12$	Air density $\rho = 1.14$	Air density $\rho = 1.16$	Air density $\rho = 1.18$	Air density $\rho = 1.2$	Air density $\rho = 1.225$
	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³	kg/m ³
3.0	43	45	47	49	51	53	54	56	58	60	62
3.5	107	110	114	117	120	123	126	129	132	135	139
4.0	187	192	197	201	206	211	215	220	224	229	235
4.5	283	289	296	302	309	316	322	329	335	342	350
5.0	397	406	415	424	432	441	450	459	467	476	487
5.5	536	547	559	570	582	593	605	616	628	639	654
6.0	698	713	728	742	757	772	786	801	816	830	849
6.5	888	907	925	944	962	980	998	1016	1034	1052	1074
7.0	1108	1131	1153	1176	1198	1220	1242	1265	1287	1309	1337
7.5	1354	1381	1408	1435	1462	1487	1512	1537	1562	1587	1617
8.0	1627	1658	1690	1721	1753	1781	1809	1836	1864	1892	1924
8.5	1918	1954	1991	2027	2063	2092	2121	2149	2178	2207	2238
9.0	2215	2254	2294	2333	2372	2400	2428	2457	2485	2513	2544
9.5	2515	2555	2596	2636	2677	2702	2728	2753	2779	2804	2830
10.0	2791	2830	2870	2909	2948	2970	2992	3015	3037	3059	3081
10.5	3037	3072	3106	3141	3176	3194	3213	3231	3250	3268	3285
11.0	3248	3280	3313	3345	3378	3393	3408	3424	3439	3454	3467
11.5	3438	3463	3488	3513	3538	3549	3560	3570	3581	3592	3602
12.0	3563	3583	3602	3622	3641	3649	3656	3664	3671	3679	3685
12.5	3654	3668	3683	3697	3711	3716	3721	3727	3732	3737	3741
13.0	3707	3718	3729	3740	3751	3755	3759	3763	3767	3771	3774
13.5	3755	3762	3770	3777	3784	3787	3789	3792	3794	3797	3798
14.0	3783	3788	3792	3797	3802	3804	3805	3807	3808	3810	3811
14.5	3801	3804	3807	3810	3813	3814	3815	3817	3818	3819	3819
15.0	3809	3812	3814	3817	3819	3820	3821	3821	3822	3823	3823
15.5	3812	3814	3815	3817	3819	3820	3820	3821	3821	3822	3823
16.0	3817	3819	3820	3822	3823	3823	3823	3824	3824	3824	3825
16.5	3822	3823	3823	3824	3825	3825	3825	3826	3826	3826	3826
17.0	3828	3829	3829	3830	3830	3830	3830	3830	3830	3830	3830
17.5 to cutout ²	3830	3830	3830	3830	3830	3830	3830	3830	3830	3830	3830

Table 4: Calculated power curve for different values of the air density for high turbulence intensity

² For more details on the cut-out behaviour visit section 3 "Cut-Out and Re-Cut-In Wind Speeds"

3 Applicability

The power curve information provided in this document applies under the following conditions:

- The mean air density falls inside the range of air densities specified in section 2.
- The site is characterized by one of the following turbulence intensity categories.

TI Band	Lower TI bound	Upper TI bound
Low	$2.5 (0.75V_{hub} + 5.6)/V_{hub}$	$12.5 (0.75V_{hub} + 5.6)/V_{hub}$
Medium	$5 (0.75V_{hub} + 5.6)/V_{hub}$	$15 (0.75V_{hub} + 5.6)/V_{hub}$
High	$10 (0.75V_{hub} + 5.6)/V_{hub}$	$20 (0.75V_{hub} + 5.6)/V_{hub}$

Measured power curve data points for 10-minute average mean turbulence intensities that fall outside the applicable TI band would be filtered out for the purpose of power curve applicability, see Figure 1.

The reference power curve is to be selected from the power curves of the TI class having the largest number of post-filtering data points (upmost coverage). Section 6 of this document gives an example outlining how to determine the TI class with the upmost coverage. The reference power curve is referred to as the calculated power curve in GE's Technical Documentation Machine Power Performance Test. This document also outlines the data filtering requirements, which have to be met prior to determining the TI class.

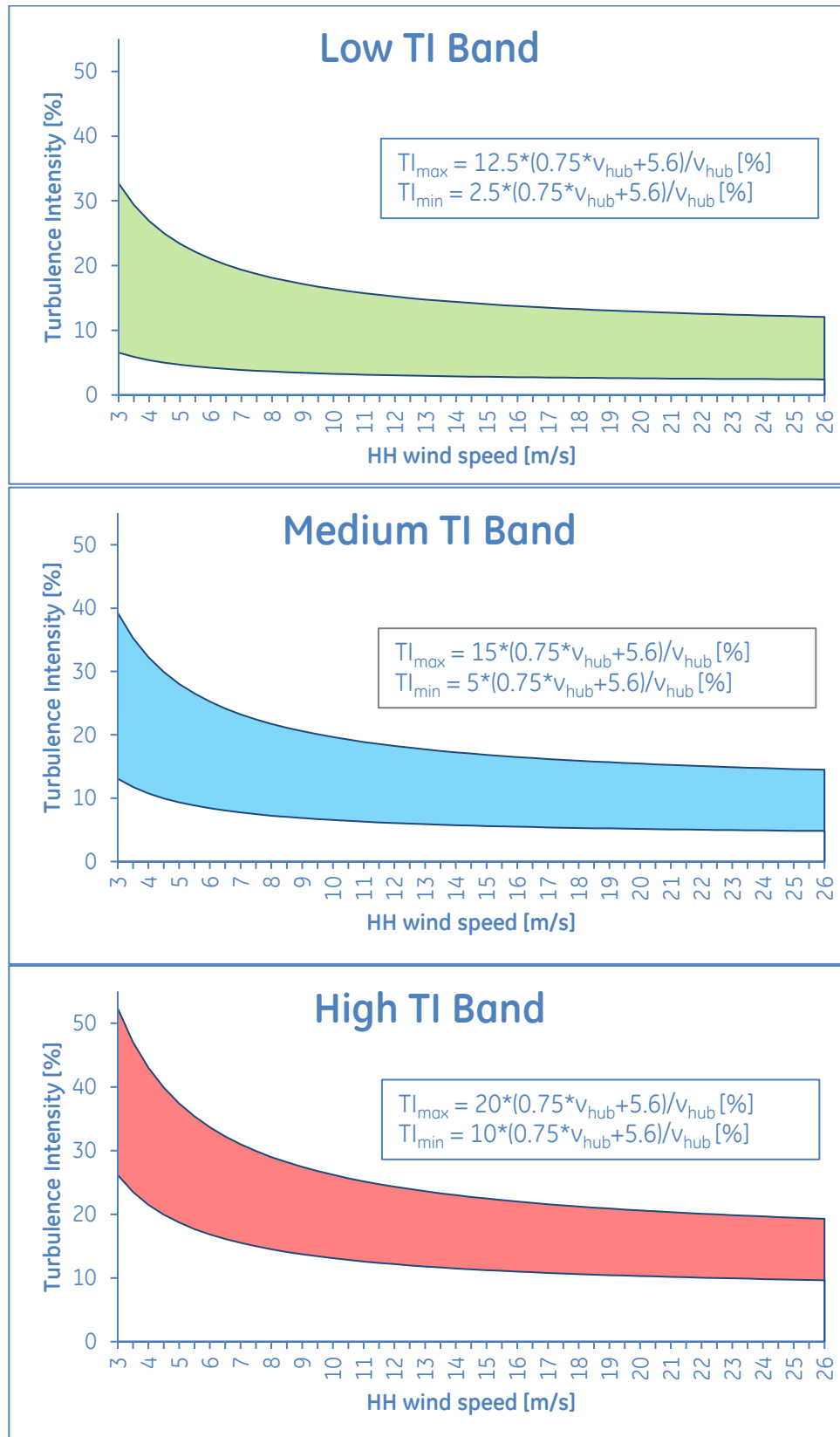


Figure 1: Applicable Turbulence Intensity Range

- Non-degraded and uncontaminated blade surfaces with no icing
- The wind turbine generator system whose power output is not being actively regulated or curtailed for any reason.
- Power shall be measured on the generator side of the transformer unless the net electric power output is referenced otherwise in this document.
- Wind-speed labels are mid-bin values; for example, the 5.0 m/s bin extends from 4.75 to 5.25 m/s.
- The flow inclination angle at the site corresponds to $-5^{\circ} < \text{flow angle} < +5^{\circ}$.
- Measured wind shear (10-minute average data) such that $-0.1 \leq \text{shear} \leq 0.5$.
- The wind turbine generator system whose power output not being curtailed due to high or low ambient temperature and/or low density. Information on the influence of cold or hot weather and high altitude operation is provided in the documents "Technical Description – Cold Weather Adaptations" and "General Description High Temperature, High Altitude Operation".

4 Cut-Out and Re-Cut-In Wind Speeds

The wind turbine generator system will shut down when the average wind speed exceeds

- 25 m/s average in a 600-second time interval.
- 30 m/s average in a 30-second time interval or
- 34 m/s average in a 3-second time interval

After the turbine has been shut-down, it will restart when the 4-minute rolling average wind speed drops below 22 m/s.

A site-specific Mechanical Loads Analysis (MLA) is required to determine the turbine suitability and whether the above-listed cut-out and re-cut-in wind speeds require adjustment to mitigate site-specific loads. If, for example, the MLA concludes that curtailment is needed to reduce loads, the MLA report will state the reduced 600-second cut-out wind speeds and the applicable wind direction sector(s). The remaining cut-out and re-cut-in wind speeds will be reduced by the same amount that the 600-second wind speed is reduced. Any reduction of the wind speeds listed in this section may be turbine location-specific. Refer to the site-specific MLA report for details.

5 Calculated Thrust Coefficient

Wind Speed at Hub Height [m/s]	Thrust Coefficient [ct]	Wind Speed at Hub Height [m/s]	Thrust Coefficient [ct]
3.0	1.01	14.5	0.21
3.5	0.95	15.0	0.19
4.0	0.91	15.5	0.17
4.5	0.88	16.0	0.16
5.0	0.84	16.5	0.14
5.5	0.82	17.0	0.13
6.0	0.81	17.5	0.12
6.5	0.81	18.0	0.11
7.0	0.81	18.5	0.10
7.5	0.80	19.0	0.10
8.0	0.79	19.5	0.09
8.5	0.76	20.0	0.08
9.0	0.71	20.5	0.08
9.5	0.65	21.0	0.07
10.0	0.59	21.5	0.07
10.5	0.54	22.0	0.07
11.0	0.48	22.5	0.06
11.5	0.43	23.0	0.06
12.0	0.38	23.5	0.06
12.5	0.33	24.0	0.05
13.0	0.30	24.5	0.05
13.5	0.26	25.0	0.05
14.0	0.23		

Table 5: Calculated thrust coefficient table; as a function of hub height wind speed

Calculated using Standard Atmospheric Conditions according to ISO 2533.

6 Guidelines for Pre-construction Energy Assessment

The following guidelines are offered for selecting power curves from this document for measurement purposes or assessment of the energy production of a new wind plant.

The air density should be the annual average air density expected at the site of the new wind plant over the life of the wind turbine.

The reference power curve for warranty purposes shall be chosen for the turbulence intensity range covering the largest fraction of data points from the measured data set of the power curve test. Figure 2 depicts one example to illustrate this guideline. The mean turbulence intensity distribution for the site is plotted over the low, medium, and high turbulence intensity ranges. As shown in Figure 2, the site has a mean turbulence intensity distribution that is most common to the low turbulence intensity range. Accordingly, the low turbulence PCs from Table 3 should be used for this example site.

For site energy assessment purposes the turbulence intensity range between low, medium, and high may also be selected based on comparing the measured 10 min averaged scatter plot in each wind speed bin expected individually to represent the conditions at the site of the new wind plant over the life of the wind turbine.

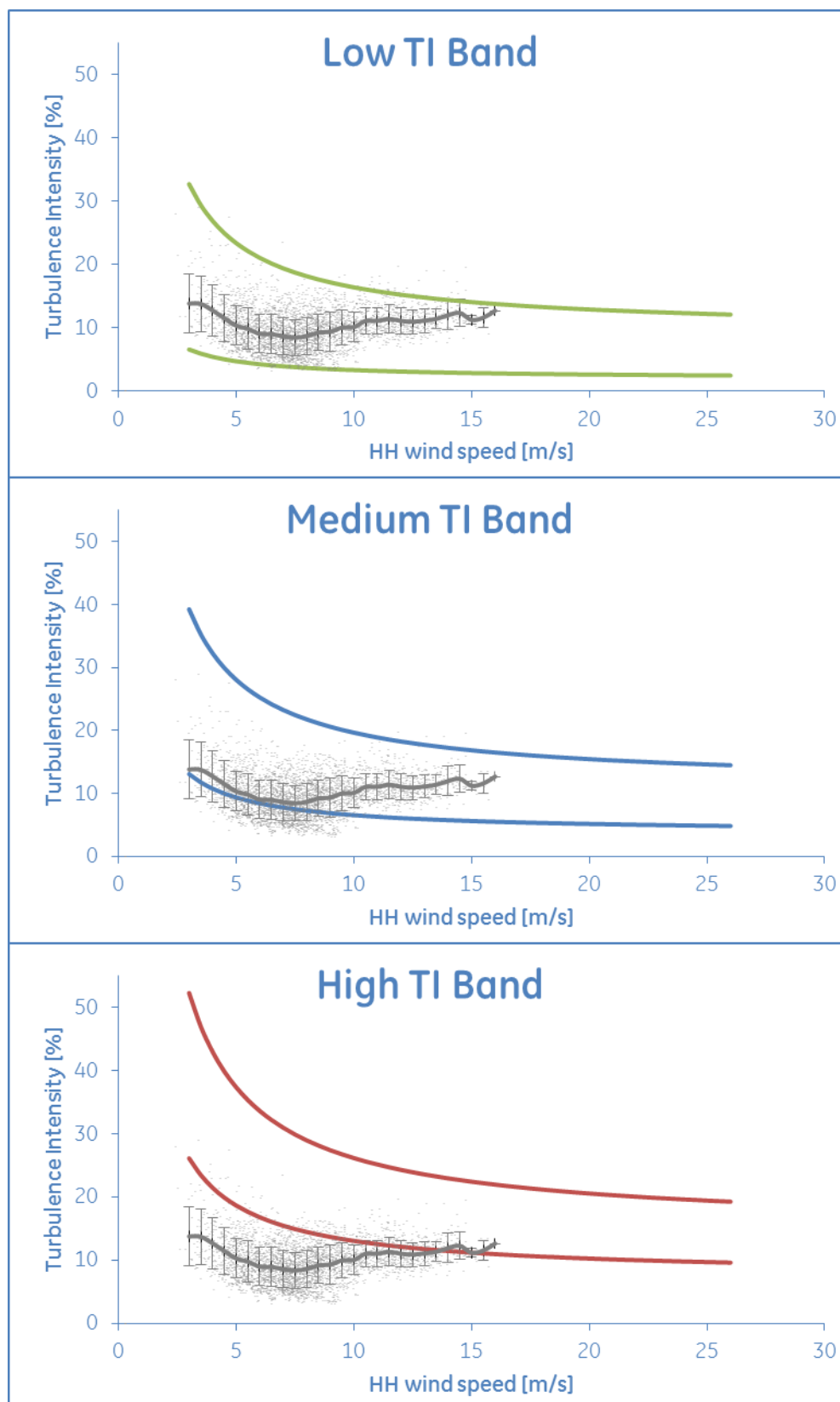


Figure 2: Example of site specific turbulence distribution with mean turbulence data and standard deviation relative to high TI (9%), medium TI (71%) and low TI (98%) band (figures in parenthesis represent the coverage rate)